

Chapter 13: Electromagnetic Induction

Mini Investigation: Electric Current from Motion?, page 587

A. Answers may vary. Sample answer:

Moving the magnet with a fast motion near the loop of wire caused a current in the wire. When the magnet was moved quickly through the loop of wire, brought quickly near the wire, or moved past the wires outside of the loop, the galvanometer showed a current in the wire.

B. Answers may vary. Sample answer:

Moving the magnet quickly through the wire was the most effective motion at producing current. The galvanometer consistently showed a significant current in the wire when the magnet was quickly moved through the loop's centre.

Section 13.1: Electromagnetic Induction

Mini Investigation: Faraday's Ring, page 588

A. Answers may vary. Sample answers:

When the primary circuit switch was closed, the galvanometer needle moved to the right, indicating that there was an electric current in the secondary circuit. The galvanometer needle then quickly returned to its original position.

B. Answers may vary. Sample answers:

When the primary circuit switch was opened, the galvanometer needle moved to the left, indicating that there was an electric current in the secondary circuit. The galvanometer needle then quickly returned to its original position.

C. Answers may vary. Sample answers:

The direction of the current in Step 3 was opposite to the direction of the current in Step 2.

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1. (a) The galvanometer would indicate a small current in the wire while the magnet is moved toward it, but would indicate no current in the wire once the magnet is placed on top of it.

(b) The galvanometer would indicate no current in the wire while the magnet is resting on top of it, but would then indicate a small current in the direction opposite to that of part (a) while the magnet is being moved away from it.

(c) The galvanometer would indicate a small current in the wire.

(d) The galvanometer would indicate a large current in the wire.

(e) The galvanometer would indicate a large current in the wire that changes direction each time the magnet changes direction.

(f) The galvanometer would indicate a large current in the wire.

2. Answers may vary. Sample answer:

To maximize the amount of induced current in a coiled conductor with a permanent magnet design, I would include a coil with a great number of loops and a strong permanent magnet. I would also have the strong permanent magnet move quickly in and out of the coil. To maximize the amount of induced current for a Faraday's ring design, I would include a great number of loops around the ring and a large amount of electric current in the primary circuit. I would also have the current in the primary circuit rapidly increase and decrease.

3. An induction cooker cannot induce an electric current in the glass, so the pot would remain at room temperature. The cooker can induce an electric current in the iron handle, so the handle would be at a very high temperature.

4. Answers may vary. Sample answer:

No, you cannot design a non-metal detector that uses electromagnetic induction because a magnetic field cannot induce current in a non-conductor, and many non-metals are non-conductors.

5. I may have to remove my belt, empty my pockets, and remove my shoes because these items contain metal that will be detected by the metal detector.

6. Answers may vary. Sample answer:

Induction charges could be used for electric toothbrushes, electric razors, cellphones, digital audio players, handheld gaming consoles, and any other portable electric devices that use batteries. Induction charging is highly useful for electric devices that are exposed to water, such as electric toothbrushes and razors, because it is safer than charging with wires. For other electric devices, the main benefit of induction charging is convenience. The main disadvantage of induction charging is that it is less efficient than direct wired contact. It has an increased resistive heating relative to direct contact. Induction charging also requires drive electronics and coils that increase the cost and complexity of manufacturing if used in a device. Whether induction charging is better than direct wired contact depends on the electric device.